The Chain Rule & Directional Derivatives

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Find the directional derivative of $f(x, y) = x^2y - 3y^2$ at the point $P(1, 2)$ in the direction of the vector $\mathbf{v} = \langle 3, 4 \rangle$.

1. Compute the gradient vector *ablaf*: $ablaf(x, y) = \langle 2xy,\ x^2 - 6y \rangle$. At $P(1, 2)$: $ablaf(1, 2) = \langle 2(1)(2),\ 1^2 - 6(2) \rangle = \langle 4,\ -11 \rangle$. 2. Normalize the direction vector $\mathbf{v}$: $||\mathbf{v}|| = \sqrt{3^2 + 4^2} = 5 \Rightarrow \mathbf{u} = \frac{\mathbf{v}}{||\mathbf{v}||} = \left\langle \frac{3}{5},\ \frac{4}{5} \right\rangle$. 3. Compute the directional derivative.

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

For $f(x, y) = -12x - 35y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

$ablaf = (-12,\ -35)$ everywhere. By Theorem 2.1, the maximum directional derivative equals $ablaf = (-12)^2 + (-35)^2 = 37$. Key Mathematical Principle & Verification: In exam revision, always verify that the final value satisfies the domain constraints, units, and original governing equation by direct back-substitution into the prompt.

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Prove that the 3D vector field $\mathbf{F} = \langle 2xy + z^2,\ x^2 + 2yz,\ 2xz + y^2 \rangle$ is conservative on $\mathbb{R}^3$, construct a potential function f satisfying $ablaf = \mathbf{F}$, and evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(0, 0, 0)$ to $(1, 2, 3)$.

1. Verify $abla \times \mathbf{F} = \mathbf{0}$: $abla \times \mathbf{F} = \left|\mathbf{i}, \mathbf{j}, \mathbf{k};\ \partial_x, \partial_y, \partial_z;\ 2xy + z^2, x^2 + 2yz, 2xz + y^2\right| = \langle 2y - 2y,\ 2z - 2z,\ 2x - 2x \rangle = \mathbf{0}$. Since $\mathbb{R}^3$ is simply connected, $\mathbf{F}$ is conservative. 2. Reconstruct the potential function:Integrate $f_x = 2xy + z^2 \Rightarrow f(x, y, z) = x^2y + xz^2 + g(y, z)$.Differentiate with respect to y.

4 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Evaluate the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ for the vector field $\mathbf{F} = \langle y,\ -x, z \rangle$ along the circular helix $\mathbf{r}(t) = \langle \cos t, \sin t, t \rangle$ for $0 \le t \le 2\pi$.

1. Velocity vector: $\mathbf{r}'(t) = \langle -\sin t,\ \cos t,\ 1 \rangle$. 2. Evaluate $\mathbf{F}$ along the curve: With $x = \cos t, y = \sin t, z = t$: $\mathbf{F}(\mathbf{r}(t)) = \langle \sin t,\ -\cos t,\ t \rangle$. 3. Compute dot product: $\mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) = (\sin t)(-\sin t) + (-\cos t)(\cos t) + (t)(1) = -\sin^2 t - \cos^2 t + t = -1 + t$. 4. Integrate.

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5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the vector field $\mathbf{F}(x, y) = \langle 2x + y, x - 3y \rangle$. What is the value of the vector field at the point $(3, -1)$?

To find the value of the vector field $\mathbf{F}(x, y) = \langle 2x + y, x - 3y \rangle$ at the point $(3, -1)$, we substitute $x = 3$ and $y = -1$ into the components of the vector field. The first component is $2x + y = 2(3) + (-1) = 6 - 1 = 5$. The second component is $x - 3y = 3 - 3(-1) = 3 + 3 = 6$. Therefore, the vector field at $(3, -1)$ is $\langle 5, 6 \rangle$. Final Answer: $\langle 7, 0 \rangle$

6 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Maximize $f(x, y) = 5xy$ subject to $x + y = -10$. Find the maximum value.

$g(x, y) = x + y$, $ablag = (1, 1)$. $ablaf = (5y, 5x)$; setting $ablaf = \lambda ablag$ gives $5y = \lambda = 5x \Rightarrow x = y$. With $x + y = -10$: $x = y = -5$, so $f = 125$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the total mass and the z-coordinate of the center of mass of the upper solid hemisphere $x^2 + y^2 + z^2 \le R^2$, $z \ge 0$, whose density is proportional to the distance from the origin: $\delta(\rho, \phi, \theta) = k\rho$ for constant $k > 0$.

1. Coordinate Setup: In spherical coordinates: $0 \le \rho \le R$, $0 \le \phi \le \frac{\pi}{2}$, $0 \le \theta \le 2\pi$, $dV = \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$. 2. Total Mass M: $M = \iiint_E (k\rho)(\rho^2 \sin\phi) \, d\rho \, d\phi \, d\theta = k\left(\int_0^{2\pi} d\theta\right)\left(\int_0^{\pi/2} \sin\phi \, d\phi\right)\left(\int_0^R \rho^3 \, d\rho\right)$. $M = k(2\pi)(1)\left(\frac{R^4}{4}\right) = \frac{\pi k R^4}{2}$. 3. Moment with respect to xy-plane M_{xy}: Since.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Compute the moment of inertia I_z about the z-axis of a solid homogeneous right circular cone of base radius R, height h, and uniform total mass M.

1. Cone Geometry & Density: The cone has vertex at the origin and base at $z = h$, so the lateral surface is $r = \frac{R}{h} z \Rightarrow z = \frac{h}{R} r$. Total volume $V = \frac{1}{3}\pi R^2 h$, so density $\delta_0 = \frac{3M}{\pi R^2 h}$. 2. Cylindrical Integral Setup for $I_z = \iiint_E r^2 \delta_0 \, dV$: Bounds: $0 \le \theta \le 2\pi$, $0 \le r \le R$, $\frac{h}{R} r \le z \le h$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the volume and the z-coordinate of the centroid $\bar{z}$ of the solid region bounded below by the xy-plane $z = 0$ and above by the paraboloid $z = 4 - x^2 - y^2$, assuming uniform density $\delta_0 = 1$.

1. Cylindrical Coordinate Setup: The paraboloid intersects $z = 0$ at $r^2 = 4 \Rightarrow r = 2$. Bounds: $0 \le \theta \le 2\pi$, $0 \le r \le 2$, $0 \le z \le 4 - r^2$, with $dV = r \, dz \, dr \, d\theta$. 2. Volume: $V = \int_0^{2\pi} \int_0^2 \int_0^{4-r^2} r \, dz \, dr \, d\theta = 2\pi \int_0^2 r(4 - r^2) \, dr = 2\pi [2r^2 - \frac{r^4}{4}]_0^2 = 2\pi(8 - 4) = 8\pi$. 3. Moment M_{xy}.

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10 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Evaluate the Cartesian triple integral $\iiint_B xyz^2\,dV$ over the rectangular box $B = [0,1] \times [-1,2] \times [0,3]$.

Since the region is a Cartesian box and the integrand is completely separable into univariate functions $f(x,y,z) = x \cdot y \cdot z^2$, Fubini's Theorem allows complete factoring: $\iiint_B xyz^2\,dV = \left(\int_0^1 x\,dx\right)\left(\int_{-1}^2 y\,dy\right)\left(\int_0^3 z^2\,dz\right)$. Compute each single-variable factor: $\int_0^1 x\,dx = [\frac{x^2}{2}]_0^1 = \frac{1}{2}$. $\int_{-1}^2 y\,dy = [\frac{y^2}{2}]_{-1}^2 = \frac{4-1}{2} = \frac{3}{2}$..

11 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Evaluate the double integral $\iint_R x\cos(x+y)\,dA$ over the rectangular region $R = [0,\pi] \times [0,\pi/2]$.

1. Set up the iterated integral: $\int_0^{\pi}\int_0^{\pi/2} x\cos(x+y)\,dy\,dx$. 2. Integrate with respect to y: $\int_0^{\pi/2}\cos(x+y)\,dy = [\sin(x+y)]_0^{\pi/2} = \sin(x+\pi/2) - \sin x = \cos x - \sin x$. 3. Integrate with respect to x using integration by parts: $\int_0^{\pi} x(\cos x - \sin x)\,dx = \int_0^{\pi} x\cos x\,dx - \int_0^{\pi} x\sin x\,dx$. $\int_0^{\pi} x\cos x\,dx = [x\sin x]_0^{\pi} - \int_0^{\pi}\sin x\,dx = 0 - [-\cos x]_0^{\pi} = -(1-(-1)) = -2$..

12 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Classify the critical points $(0,0)$ and $(1,1)$ of $f(x,y) = x^3 + y^3 - 3xy$ using the Second Derivative Test and Hessian matrix.

1. Compute the second partial derivatives: $f_{xx} = 6x, \quad f_{yy} = 6y, \quad f_{xy} = -3$. The Hessian determinant discriminant is: $D(x,y) = f_{xx}f_{yy} - (f_{xy})^2 = (6x)(6y) - (-3)^2 = 36xy - 9$. 2. Evaluate at $(0,0)$: $D(0,0) = 36(0) - 9 = -9 < 0$. Since $D < 0$, $(0,0)$ is a saddle point. 3. Evaluate at $(1,1)$: $D(1,1) = 36(1)(1) - 9 = 27 > 0, \quad f_{xx}(1,1) = 6(1) = 6 > 0$. Since $D > 0$ and $f_{xx} > 0$, $(1,1)$ is a local minimum with value $f(1,1) = 1 + 1 - 3 = -1$.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Find the surface area of the paraboloid cap $z = x^2 + y^2$ that lies below the horizontal plane $z = 4$.

1. Surface Area Differential for a Graph $z = g(x,y)$: $g_x = 2x, \quad g_y = 2y \Rightarrow dS = \sqrt{1 + (2x)^2 + (2y)^2}\,dA = \sqrt{1 + 4(x^2+y^2)}\,dA$. 2. Projection onto xy-plane: The intersection $x^2 + y^2 = 4$ is a disk D of radius 2. In polar coordinates: $0 \le r \le 2$, $0 \le \theta \le 2\pi$, and $dA = r\,dr\,d\theta$. 3. Evaluate the surface integral.

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Prove that the limit $\lim_{(x,y)\to(0,0)} \frac{x^2y}{x^4+y^2}$ does not exist by using the Two-Path Test, comparing straight-line approaches with parabolic approaches.

1. Approach along linear paths $y = mx$: $\lim_{x\to0} \frac{x^2(mx)}{x^4+(mx)^2} = \lim_{x\to0} \frac{mx^3}{x^2(x^2+m^2)} = \lim_{x\to0} \frac{mx}{x^2+m^2} = \frac{0}{m^2} = 0 \quad (\text{for } m \neq 0)$. Along $y = 0$ (the x-axis), $\lim_{x\to0} \frac{0}{x^4} = 0$. Along all straight lines through the origin, the limit is 0. 2. Approach along parabolic paths $y = kx^2$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives

A solid region E is defined by the inequalities $x^2 + y^2 \le 16$, $0 \le z \le 5$. The volume of this region can be computed using a triple integral. In cylindrical coordinates, the region E is described by $0 \le r \le R$, $0 \le \theta \le 2\pi$, and $0 \le z \le H$. What is the value of R?

The given region is a cylinder with radius R_{cyl} and height H_{cyl}. The inequality $x^2 + y^2 \le 16$ describes a circle of radius 4 centered at the origin in the xy-plane. When converting to cylindrical coordinates, $x^2 + y^2$ becomes r^2. Thus, the inequality $r^2 \le 16$ implies $r \le 4$. Therefore, in cylindrical coordinates, the radial component of the region is described by $0 \le r \le 4$. The question states that the region is described by $0 \le r \le R$, so $R = 4$. The inequality.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers

Find the directional derivative of $f(x, y, z) = x^2y + yz^2$ at $P(1, 2, -1)$ in the direction of $\mathbf{v} = \langle 2, -1, 2 \rangle$.

Step 1: Compute the gradient $ablaf$: $ablaf = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle = \langle 2xy,\ x^2 + z^2,\ 2yz \rangle$. Step 2: Evaluate $ablaf$ at $P(1, 2, -1)$: $ablaf(1, 2, -1) = \langle 2(1)(2),\ 1^2 + (-1)^2,\ 2(2)(-1) \rangle = \langle 4, 2, -4 \rangle$. Step 3: Normalize direction vector $\mathbf{v}$: $||\mathbf{v}|| = \sqrt{2^2 + (-1)^2 + 2^2} = \sqrt{4 + 1 + 4} = 3$. .

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 4.01 Double Integrals

Let $z = x^2y - y^3$, where $x(t) = \cos(2t)$ and $y(t) = \sin(2t)$. Use the multivariable chain rule to find $\frac{dz}{dt}$ when $t = 0$.

Step 1: Compute partial derivatives of z: $\frac{\partial z}{\partial x} = 2xy$, $\frac{\partial z}{\partial y} = x^2 - 3y^2$. Step 2: Compute derivatives of coordinate functions: $\frac{dx}{dt} = -2\sin(2t)$, $\frac{dy}{dt} = 2\cos(2t)$. Step 3: Evaluate at $t = 0$: $x(0) = \cos(0) = 1$, $y(0) = \sin(0) = 0$. $\frac{\partial z}{\partial x}(1, 0) = 0$, $\frac{\partial z}{\partial y}(1, 0) = 1^2 - 0 = 1$. $\frac{dx}{dt}(0) = 0$, $\frac{dy}{dt}(0) = 2$. Step 4: Apply the chain rule.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives

[The Multivariable Chain Rule, Directional Derivatives & Gradient Geometry · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

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19 **[Challenging]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)** • **Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers**

Given the multivariable function $f(x, y) = x^3y^2 - 4xy^3 + 5x^2 - 7y + 2$, find the first-order partial derivatives $f_x(x, y)$ and $f_y(x, y)$, and evaluate $f_x(1, 2)$.

Step 1: Differentiate with respect to x (holding y constant):

$f_x(x, y) = \frac{\partial}{\partial x}(x^3y^2 - 4xy^3 + 5x^2 - 7y + 2) = 3x^2y^2 - 4y^3 + 10x.$ Step 2: Differentiate with respect to y (holding x constant): $f_y(x, y) = \frac{\partial}{\partial y}(x^3y^2 - 4xy^3 + 5x^2 - 7y + 2) = 2x^3y - 12xy^2 - 7.$ Step 3: Evaluate $f_x(1, 2)$:

$f_x(1, 2) = 3(1)^2(2)^2 - 4(2)^3 + 10(1) = 3(4) - 4(8) + 10 = 12 - 32 + 10 = -10.$

20 **[Challenging]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)** • **Cross-Lesson: 4.01 Double Integrals**

Find an equation of the tangent plane to the surface $z = x^2 + 3y^2 - 2x + 4y$ at the point $P(2, 1, 9)$.

Step 1: Compute partial derivatives: $f_x(x, y) = 2x - 2, \quad f_y(x, y) = 6y + 4.$ Step 2: Evaluate at $(x_0, y_0) = (2, 1)$: $f_x(2, 1) = 2(2) - 2 = 2, \quad f_y(2, 1) = 6(1) + 4 = 10.$ Step 3: Apply the tangent plane formula: $z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$ $z - 9 = 2(x - 2) + 10(y - 1)$ $z - 9 = 2x - 4 + 10y - 10 \Rightarrow 2x + 10y - z = 5.$

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 **[Foundational]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

If $z = f(x, y)$ where $x = g(t)$ and $y = h(t)$, what is $\frac{dz}{dt}$?

a. $\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$ ✓

b. $\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dy}{dt} + \frac{\partial z}{\partial y}\frac{dx}{dt}$

c. $\frac{dz}{dt} = \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y}$

d. $\frac{dz}{dt} = \frac{dx}{dt}\frac{dy}{dt}$

By the multivariable chain rule, rates of change sum across all intermediate paths: $\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$.

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22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the gradient vector $\nabla f(x, y)$ of a scalar function $f(x, y)$?

a. $\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right]$ ✓

b. $\nabla f = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}$

c. $\nabla f = \left[\frac{\partial^2 f}{\partial x^2}, \frac{\partial^2 f}{\partial y^2}\right]$

d. $\nabla f = \sqrt{f_x^2 + f_y^2}$

The gradient of a multivariable function is the vector consisting of its first-order partial derivatives: $\nabla f = [f_x, f_y]$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Compute the gradient $\nabla f(2, -1)$ for $f(x, y) = 3x^2y + 2y^2$.

a. $[-12, 8]$ ✓

b. $[12, -8]$

c. $[-12, -4]$

d. $[6, 4]$

$f_x = 6xy \Rightarrow f_x(2, -1) = 6(2)(-1) = -12$. $f_y = 3x^2 + 4y \Rightarrow f_y(2, -1) = 3(4) + 4(-1) = 12 - 4 = 8$. Gradient is $[-12, 8]$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the geometric relationship between the gradient $\nabla f(x_0, y_0)$ and the level curve $f(x, y) = c$ passing through (x_0, y_0)?

a. $\nabla f(x_0, y_0)$ **is perpendicular (orthogonal) to the tangent line of the level curve** ✓

b. $\nabla f(x_0, y_0)$ is parallel to the tangent line

c. $\nabla f(x_0, y_0)$ has zero dot product with the normal vector

d. $\nabla f(x_0, y_0)$ points along the curve

The gradient vector at any point is always perpendicular to the level curve (or level surface) passing through that point.

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)
What is the formula for the directional derivative $D_{\mathbf{u}}f$ in the direction of a unit vector $\mathbf{u}$?

a. $D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$ ✓

b. $D_{\mathbf{u}}f = \nabla f \times \mathbf{u}$

c. $D_{\mathbf{u}}f = ||\nabla f|| ||\mathbf{u}||$

d. $D_{\mathbf{u}}f = \frac{\nabla f \cdot \mathbf{u}}{||\nabla f||}$

The rate of change of f in the direction of unit vector $\mathbf{u}$ is the dot product of the gradient with the unit vector: $D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)
Why must $\mathbf{u}$ be normalized to a unit vector before computing $D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$?

a. **To ensure the derivative measures rate of change per unit of geometric distance** ✓

b. Because dot products cannot be computed with non-unit vectors

c. To avoid division by zero

d. Because the gradient has length 1

Normalizing $\mathbf{u}$ ensures that the directional derivative accurately reflects the instantaneous rate of change per unit distance along that direction.

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)
In which direction does a differentiable function f increase most rapidly at a point?

a. **In the direction of the gradient vector** ∇f ✓

b. In the direction opposite to ∇f (i.e. $-\nabla f$)

c. Orthogonal to the gradient vector

d. Along the tangent to the level curve

Since $D_{\mathbf{u}}f = ||\nabla f|| \cos\theta$, the maximum rate occurs when $\cos\theta = 1$ ($\theta = 0$), pointing in the exact direction of ∇f.

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the maximum value of the directional derivative $D_{\mathbf{u}}f$ at a point?

a. $||\nabla f||$ ✓

b. $||\nabla f||^2$

c. 0

d. 1

When $\mathbf{u}$ points in the direction of ∇f, $D_{\mathbf{u}}f = \nabla f \cdot \frac{\nabla f}{||\nabla f||} = ||\nabla f||$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In which direction is the rate of change of f equal to zero?

a. **In any direction tangent to the level curve (orthogonal to ∇f)** ✓

b. In the direction of $-\nabla f$

c. In the direction of ∇f

d. Nowhere, a differentiable function always changes

When $\mathbf{u}$ is orthogonal to ∇f, $\cos\theta = 0$, so $D_{\mathbf{u}}f = 0$, which occurs along tangent directions to level curves.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For $z = -3xy$ with $x = t^2$, $y = t$, find dz/dt at $t = 3$ using the chain rule.

a. -27

b. 81

c. **-81** ✓

d. -78

$z_x = -3y$, $z_y = -3x$, $x'(t) = 2t$, $y'(t) = 1$. $dz/dt = z_x x' + z_y y' = -9t^2$, so at $t = 3$: $dz/dt = -81$.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Chain rule with two paths: $z = f(x, y)$, $x = r\cos\theta$, $y = r\sin\theta$. $\partial z/\partial r =$

a. $f_x\cos\theta + f_y\sin\theta$ ✓

b. f_r

c. $f_x r + f_y r$

d. $f_x\sin\theta - f_y\cos\theta$

Chain rule: $z_r = f_x x_r + f_y y_r = f_x\cos\theta + f_y\sin\theta$.

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32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Consider the vector field $\mathbf{F}(x, y) = \langle 2x, 3y \rangle$. Which of the following options best represents the vector field at the point (1, 1)?

a. $\langle 2, 3 \rangle$ ✓

b. $\langle 1, 1 \rangle$

c. $\langle 4, 6 \rangle$

d. $\langle 2, 6 \rangle$

To find the representation of the vector field $\mathbf{F}(x, y) = \langle 2x, 3y \rangle$ at a specific point, we substitute the coordinates of the point into the components of the vector field. At the point (1, 1), we have: $x = 1$ and $y = 1$. Substituting these values into $\mathbf{F}(x, y)$ gives: $\mathbf{F}(1, 1) = \langle 2(1), 3(1) \rangle = \langle 2, 3 \rangle$. Therefore, the vector field at the point (1, 1) is $\langle 2, 3 \rangle$. Option B is incorrect as it represents the position vector at (1,1)..

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the region in the first octant bounded by the coordinate planes and the plane $x + y + z = 4$. If the density of the region is given by $\rho(x, y, z) = 3xy$, set up the triple integral for the mass of this region in Cartesian coordinates.

a. $\int_0^4 \int_0^{4-x} \int_0^{4-x-y} 3xy \, dz \, dy \, dx$ ✓

b. $\int_0^4 \int_0^{4-y} \int_0^{4-y-z} 3xy \, dx \, dz \, dy$

c. $\int_0^4 \int_0^{4-x} \int_0^{4-x-y} 3 \, dz \, dy \, dx$

d. $\int_0^4 \int_0^{4-z} \int_0^{4-z-y} 3xy \, dx \, dy \, dz$

The region is in the first octant, bounded by $x = 0, y = 0, z = 0$ and $x + y + z = 4$. We can express the bounds for z in terms of x and y from the plane equation: $z = 4 - x - y$. Since it's in the first octant, z goes from 0 to $4 - x - y$. The projection of this region onto the xy-plane is a triangle bounded by $x = 0, y = 0$ and $x + y = 4$. Thus, y ranges from 0 to $4 - x$, and x ranges from 0 to 4. The density is $\rho(x, y, z) = 3xy$. Therefore, the triple integral for the mass is.

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the surface S which is the portion of the plane $z = 2x + 3y + 1$ lying above the rectangular region $R = [0, 1] \times [0, 2]$ in the xy-plane. Calculate the surface area of S.

a. $\sqrt{14}$

b. $2\sqrt{14}$ ✓

c. $3\sqrt{14}$

d. $4\sqrt{14}$

The surface area of a surface $z = g(x, y)$ above a region R is given by the integral $\iint_R \sqrt{1 + (g_x)^2 + (g_y)^2}\, dA$. Here, $g(x, y) = 2x + 3y + 1$. The partial derivatives are $g_x = 2$ and $g_y = 3$. Therefore, $\sqrt{1 + (g_x)^2 + (g_y)^2} = \sqrt{1 + 2^2 + 3^2} = \sqrt{1 + 4 + 9} = \sqrt{14}$. The integral becomes $\iint_R \sqrt{14}\, dA$, where $R = [0, 1] \times [0, 2]$. This evaluates to $\sqrt{14} \times \text{Area}(R)$. The area of the rectangle R is $(1 - 0) \times (2 - 0) = 2$. So, the sur.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Consider the function $f(x, y) = x^2 + y^2$. Find the value of $f_{xx}(2, 3)$.

a. **2** ✓

b. 4

c. 6

d. 0

First, we find the first partial derivative of $f(x, y)$ with respect to x: $f_x(x, y) = \frac{\partial}{\partial x}(x^2 + y^2) = 2x$. Next, we find the second partial derivative of $f(x, y)$ with respect to x (which is f_{xx}): $f_{xx}(x, y) = \frac{\partial}{\partial x}(f_x(x, y)) = \frac{\partial}{\partial x}(2x) = 2$. Finally, we evaluate $f_{xx}(x, y)$ at the point $(2, 3)$. Since $f_{xx}(x, y)$ is a constant, its value at any point is 2. Therefore, $f_{xx}(2, 3) = 2$.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Consider the vector field $\mathbf{F}(x, y) = \langle 3x, 2y \rangle$. Evaluate the vector field at the point $(2, 4)$.

a. $\langle 6, 8 \rangle$ ✓

b. $\langle 8, 6 \rangle$

c. $\langle 5, 6 \rangle$

d. $\langle 6, 5 \rangle$

The vector field is given by $\mathbf{F}(x, y) = \langle 3x, 2y \rangle$. To evaluate the vector field at the point $(2, 4)$, we substitute $x = 2$ and $y = 4$ into the components of the vector field. First component: $3x = 3(2) = 6$. Second component: $2y = 2(4) = 8$. Therefore, $\mathbf{F}(2, 4) = \langle 6, 8 \rangle$.

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the flux $\iint_S \mathbf{F} \cdot d\mathbf{S}$ of $\mathbf{F} = \langle 0, 0, z \rangle$ across the upper hemisphere $x^2 + y^2 + z^2 = 1,\ z \geq 0$, with outward normal.

a. $\frac{2\pi}{3}$ ✓

b. $\frac{4\pi}{3}$

c. $-\frac{2\pi}{3}$

d. $\frac{\pi}{3}$

With $\mathbf{F} \cdot \mathbf{n} = z^2/1$ on the sphere, integrating in spherical coordinates gives $\frac{2\pi}{3}$ (equivalently, half the flux $\frac{4}{3}\pi a^3$ through the whole sphere).

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives

Compute $\iint_D 9(x^2 + y^2)\, dA$ where D is the disk $x^2 + y^2 \leq 49$.

a. 2058π

b. 14406π

c. 21609π/4

d. **21609π/2** ✓

In polar coordinates, D is $0 \leq r \leq 7$, $0 \leq \theta \leq 2\pi$: $\iint_D 9(x^2 + y^2)\, dA = 9\int_0^{2\pi}\int_0^7 r^3\, dr\, d\theta = 9\int_0^{2\pi} \frac{7^4}{4} d\theta = 21609\pi/2$.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers

Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F} = \langle y^2, 2x \rangle$ along the segment from $(-2,\ -2)$ to $(-2,\ 3)$.

a. 20

b. -19

c. -40

d. -20 ✓

Parametrize $\mathbf{r}(t) = \langle -2, 5t - 2 \rangle,\ 0 \leq t \leq 1$. Then $\int_0^1 \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t)\, dt = \int_0^1 (-20)\, dt = -20$.

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.01 Double Integrals

Use Stokes' theorem to find the circulation $\oint_C \mathbf{F} \cdot d\mathbf{r}$ of $\mathbf{F} = \langle -2y, 2x, y \rangle$ around the circle $x^2 + y^2 = 1$ in the plane $z = 0$, oriented counterclockwise seen from above.

a. -4π

b. 2π

c. 8π

d. 4π ✓

$abla \times \mathbf{F} = \langle 1, 0, 4 \rangle$. Taking the flat surface inside C with $\mathbf{n} = \mathbf{k}$, $\iint_S (abla \times \mathbf{F}) \cdot \mathbf{k}\, dS = 4\pi$.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives

Use Stokes' theorem to find the circulation $\oint_C \mathbf{F} \cdot d\mathbf{r}$ of $\mathbf{F} = \langle -y, x, 0 \rangle$ around the boundary of the square $[0, 2] \times [0, 2]$ in the plane $z = 0$, oriented counterclockwise seen from above.

a. 8 ✓

b. -8

c. 4

d. 16

$abla \times \mathbf{F} = \langle 0, 0, 2 \rangle$. Taking the flat surface inside C with $\mathbf{n} = \mathbf{k}$, $\iint_S (abla \times \mathbf{F}) \cdot \mathbf{k}\, dS = 8$.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers

Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F} = \langle xy, x \rangle$ along the parabola $y = x^2$ from $(0,\ 0)$ to $(1,\ 1)$.

a. $-\frac{11}{12}$

b. $\frac{11}{6}$

c. $\frac{23}{12}$

d. $\frac{11}{12}$ ✓

Substitute the parametrization: $\mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) = t^2(t + 2)$. Integrating over $[0, 1]$ gives $\frac{11}{12}$.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 4.01 Double Integrals

Which vector field is conservative on all of R^2?

a. $\langle e^y, xe^y \rangle$ ✓

b. $\langle xy, x \rangle$

c. $\langle x + y,\ -x + y \rangle$

d. $\langle x^2 y, xy^2 \rangle$

On R^2, $\langle P, Q \rangle$ is conservative iff $P_y = Q_x$. For $\langle e^y, xe^y \rangle$: $P_y = e^y = Q_x$; each other field has $P_y eQ_x$.

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives

Evaluate $\iiint_B (x^2+y^2+z^2)\,dV$ where B is the ball $x^2+y^2+z^2 \le 4$.

a. $\frac{128\pi}{5}$ ✓

b. $\frac{128\pi}{3}$

c. $\frac{64\pi}{5}$

d. $\frac{256\pi}{5}$

In spherical coordinates the integrand is ρ^2, so $\int_0^{2\pi}\int_0^{\pi}\int_0^{2}\rho^4\sin\phi\,d\rho\,d\phi\,d\theta = \frac{128\pi}{5}$.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers

$\mathbf{F} = \langle ye^x, e^x\rangle$ is conservative. Evaluate $\int_C \mathbf{F}\cdot d\mathbf{r}$ along any path from $(2,\ 0)$ to $(1,\ -1)$.

a. e

b. $-e$ ✓

c. $1-e$

d. $-2e$

A potential is $f = ye^x$. By the Fundamental Theorem of Line Integrals the value is $f(1,\ -1) - f(2,\ 0) = -e$.

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